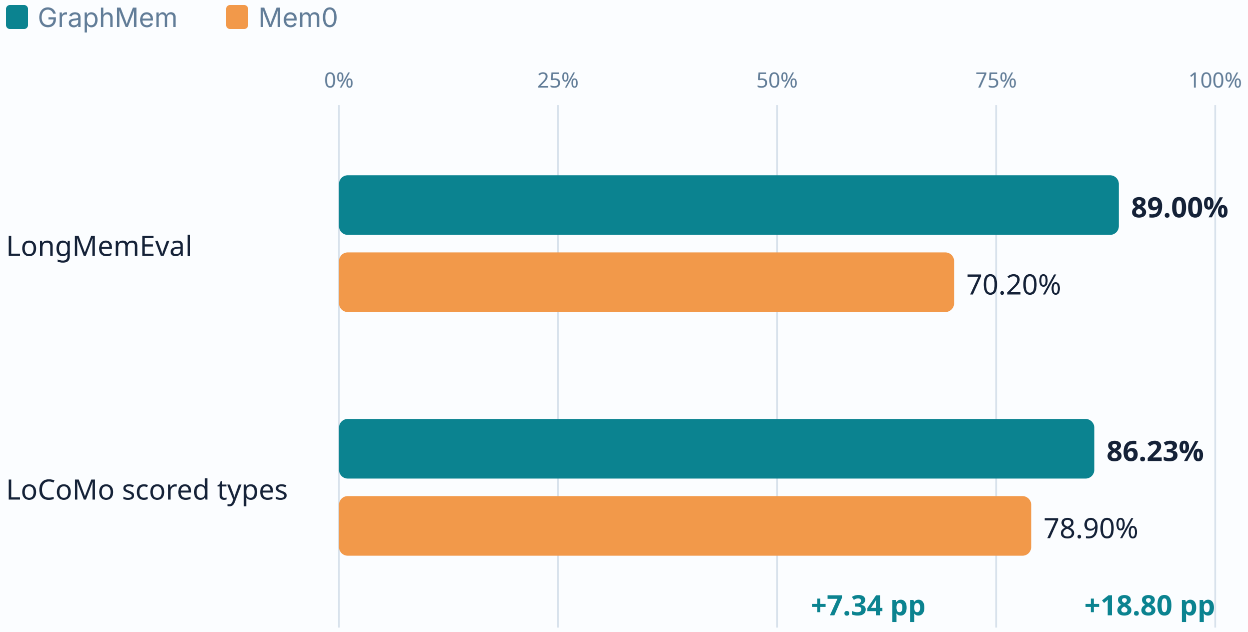
Token-Efficient, Evidence-Complete Long-Term Memory

GraphMem versus Mem0 on LongMemEval and LoCoMo — and the two architectural ideas that make retrieval both smaller and more reliable.

LongMemEval	LoCoMo · 4 scored QA types	Answer tokens / question
89.00%	**86.23%**	**7.03K · 4.41K**
+18.80 pp vs Mem0	+7.34 pp vs Mem0	LME · LoCoMo

BENCHMARK OUTCOME

GraphMem wins on both matched evaluation sets



LongMemEval paired wins

129 vs 35

Net +94 correct answers

LoCoMo paired wins

240 vs 127

Net +113 correct answers

LoCoMo IDs align across all 1,540 **Multi-hop, Temporal, Open-domain, and Single-hop** questions. Paired bootstrap 95% CI: **+4.94 to +9.68 pp**.

WHERE THE GAINS COME FROM

Graph structure matters most when evidence must be assembled

LongMemEval accuracy delta vs Mem0

Temporal reasoning	+41.35 pp
Multi-session	+20.30 pp
Assistant facts	+10.71 pp
Knowledge update	+7.69 pp
User facts	+1.43 pp
Preference	−3.33 pp

The strongest gains require **multiple roles**: two temporal endpoints, prior/current state, or evidence across sessions.

LoCoMo accuracy delta vs Mem0

Multi-hop	+17.38 pp
Single-hop	+9.75 pp
Temporal	−3.12 pp
Open-domain	−8.33 pp

Remaining weakness: Open-domain questions remain the weakest type, exposing world-knowledge integration and answer-reasoning gaps.

Raw Mem0 67.02% includes 446 Adversarial questions, which are excluded from the official scored set. The matched four-type baseline is 78.90%.

End-to-end deltas include indexing, retrieval, operators, evidence packing, and answer prompting.

TOKEN ECONOMICS

Report cost in the unit where it is actually incurred

LME build / question	LoCoMo build / conversation	LME answer / question	LoCoMo answer / question
254.5K	**132.4K**	**7,031**	**4,413**
P95 271.2K	10 builds total	P95 8,000	P95 5,055

Average token decomposition

Unit	Uncached	Cached	Output	Total
LME build / question	126,112	37,500	90,893	**254,505**
LoCoMo build / conversation	77,248	486	54,616	**132,350**
LME answer / question	5,428	1,580	23	**7,031**
LoCoMo answer / question	3,238	1,046	129	**4,413**

Why persistent memory changes the equation

```
Total cost = Build(memory) + Σ Queryᵢ(memory)
```

1. **LongMemEval:** one independent memory per question — build cannot be amortized across the benchmark.
2. **LoCoMo:** ten conversations serve 1,986 questions — build amortizes to 666 tokens per question.

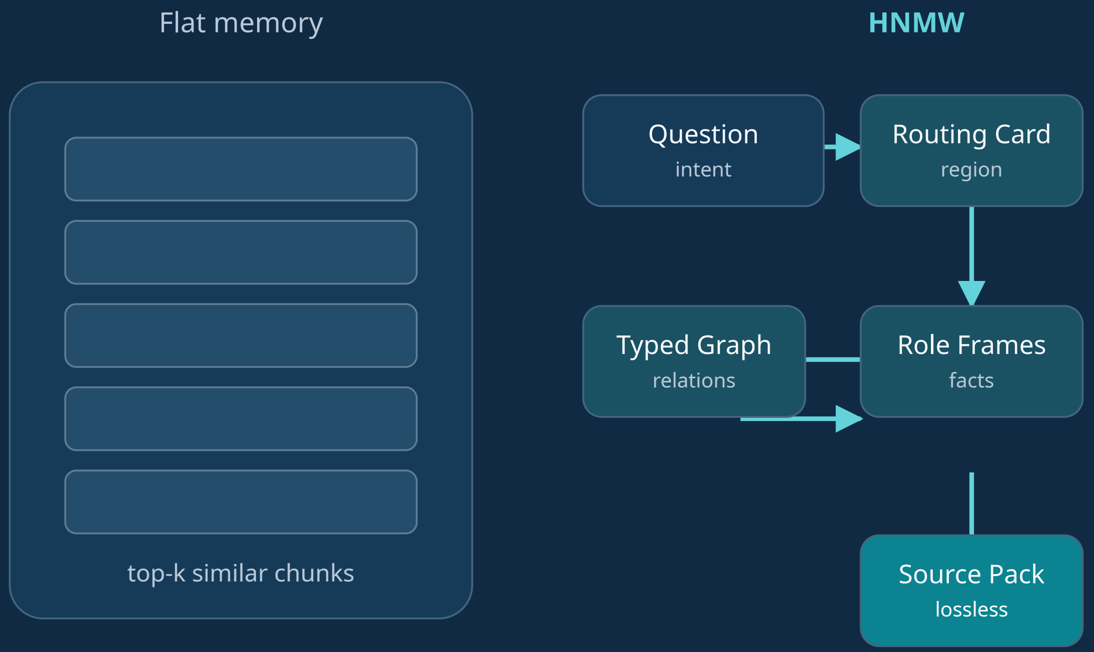
> Mem0 memory-stage usage is absent from the supplied artifacts. We report GraphMem exactly, but do not invent a cross-system token-saving percentage.

Long-term memory should be a world you can navigate — not a document you repeatedly reread

“Locate broadly, move structurally, verify losslessly.”

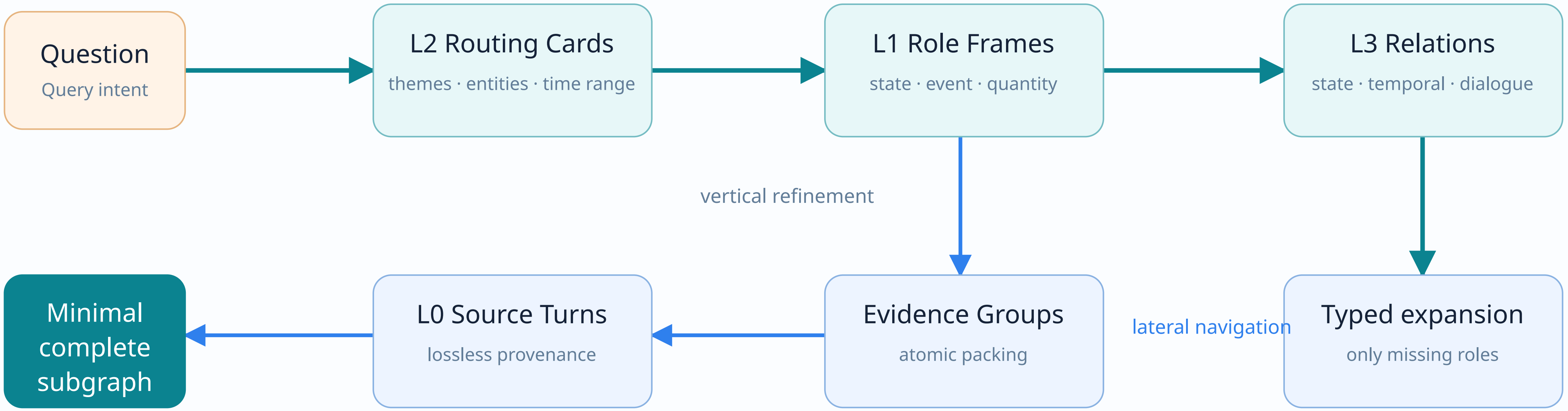
HNMW separates three jobs that flat vector memory conflates:

- **Routing** to the right memory region
- **Navigation** across state, time, event, and dialogue relations
- **Verification** against source turns



INNOVATION 1 · ARCHITECTURE

Coarse-to-fine hierarchy with lateral, typed movement



Lossless provenance

Every frame and edge resolves to original turns. Summaries route; they do not become unverified answer facts.

State-aware world model

Planned, completed, cancelled, added, and removed states coexist as ordered versions—not flattened mentions.

Bounded working set

Low-value history stays outside the prompt. Complete evidence groups enter atomically and stop expansion.

HNMW is not “retrieve a summary, then expand downward”; it supports relation-driven lateral recovery.

INNOVATION 1 · WORKED EXAMPLE

A state change becomes a navigable chain, not three competing chunks

Monday
Train to Shanghai
planned

Wednesday
Train ticket
cancelled

Wednesday
Thursday flight
booked

Question: “How am I travelling to Shanghai now?”

Flat retrieval failure mode

All three chunks are semantically relevant. A top-k prompt may over-weight the oldest explicit travel plan.

Similarity does not encode lifecycle or validity.

HNMW navigation

1. Route to the travel memory region.
2. Follow **state_transition**: planned → cancelled.
3. Recover the replacement event: flight → booked.
4. Expand only the two decisive source turns.

Result: “By flight on Thursday.” The old train plan remains auditable as superseded evidence, but cannot overwrite the latest valid state.

LME answer mean
7,031
tokens / question

LoCoMo answer mean
4,413
tokens / question

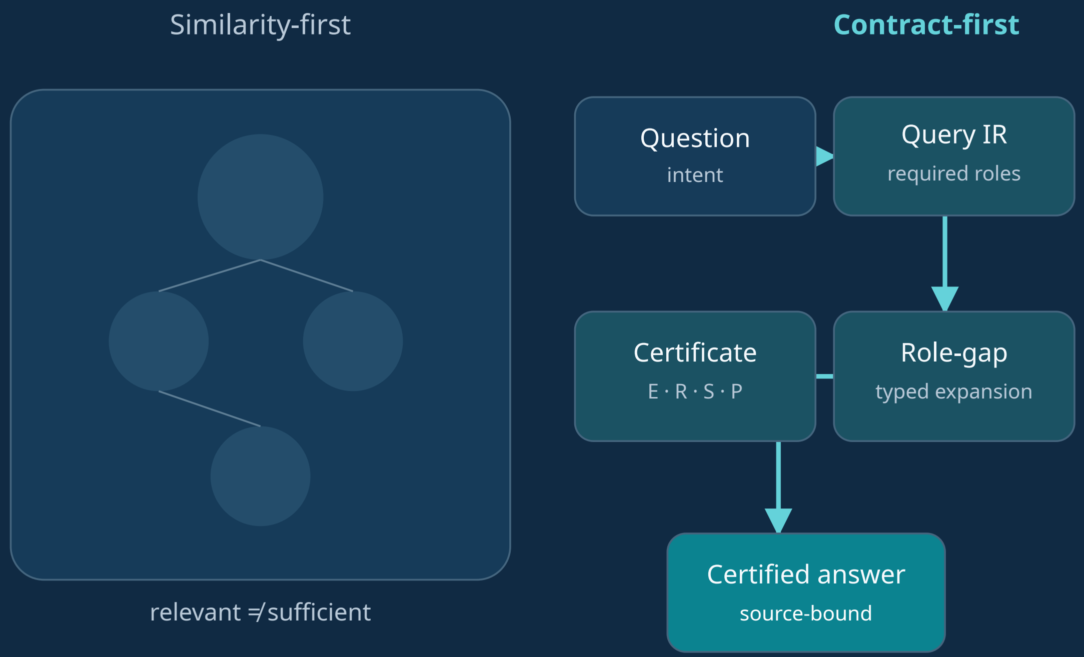
LoCoMo build reuse
1 → many
10 conversations · 1,986 queries

A question-time execution layer turns every query into an evidence contract

"At query time, compile the question into the roles a correct answer must prove."

Query IR binds all retrieval components to one plan:

target entity · owner / speaker · relation · value type · time · scope · polarity · required roles



Graph Harness runs online over a question-independent memory world

Offline · once per memory

Build the reusable memory world

1. Lossless turns preserve source authority.
2. Routing cards, role frames, state chains, and typed relations organize memory.
3. Dense, FTS/BM25, exact, and inverted indexes are persisted.

No question, gold answer, or gold session ID is read during construction.

Online · once per question

Execute a bounded evidence program

1. Compile the incoming question into deterministic Query IR.
2. Seed, diagnose missing roles, traverse allowed edges, and certify evidence.
3. Pack the smallest complete subgraph and call the answer model once.

Local Query IR, retrieval, traversal, and operators use 0 LLM tokens. An optional short planner and the final answer count toward query usage.

Compiler

Question → targets, constraints, answer algebra, required roles.

Orchestrator

Coordinates dense, BM25, exact, inverted, adjacency, and graph channels.

Verifier

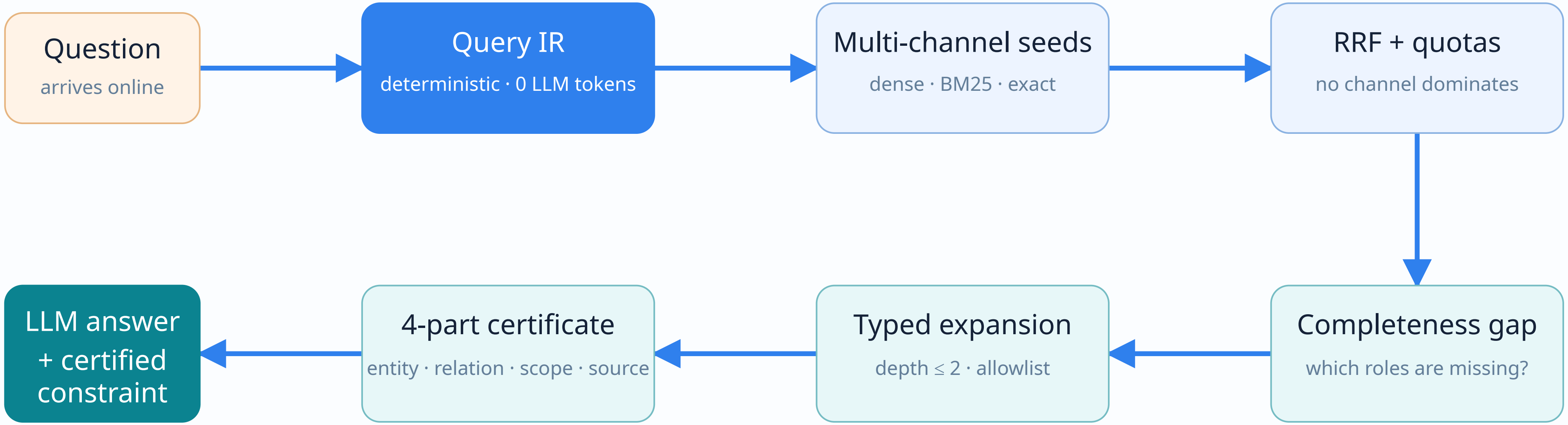
Checks entity, relation, scope, and provenance before computation.

Governor

Controls edge allowlists, depth, stopping, packing, and token budget.

INNOVATION 2 · ONLINE EXECUTION LOOP

One query contract governs retrieval, graph expansion, operators, and stopping



Stop when required roles are complete — not when top-k is exhausted

Entity

Is this the exact person, object, or event?

Relation

Does the evidence prove the requested relation?

Scope

Is time, collection, owner, and lifecycle correct?

Provenance

Can every operand resolve to source turns?

If roles remain missing, one short planner may propose aliases or roles; its output must still pass source-bound verification.

Required roles turn ambiguous retrieval into verifiable evidence assembly

Temporal comparison

“How many days earlier was Event A than Event B?”

```
roles = {A, time(A), B, time(B), source}
```

1. Retrieve A and candidate temporal facts.
2. If B is missing, expand only via **same_event / temporal_endpoint**.
3. Run date-difference only after both endpoints and sources pass certificates.

Prevents the model from inferring a difference from one endpoint.

Peer dialogue

“What gift did Speaker B buy?”

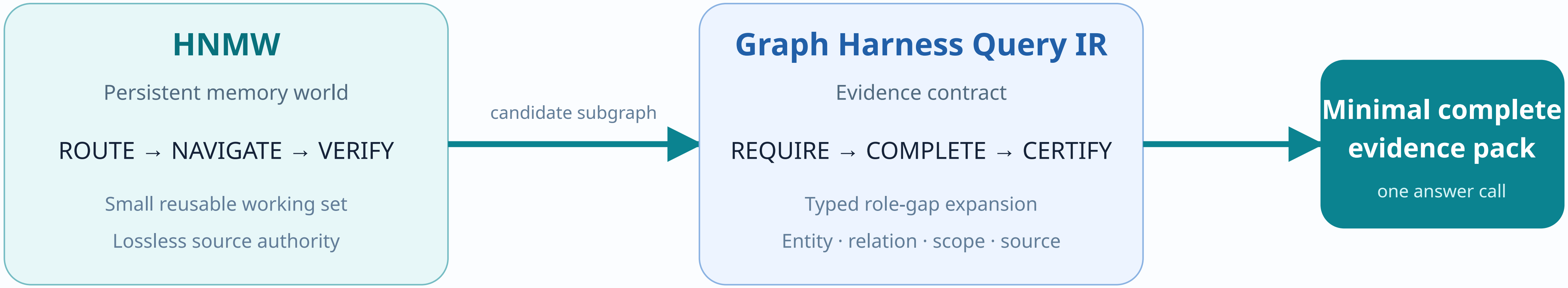
```
roles = {speaker B, purchase, object, reply, source}
```

1. Seed on “gift” and candidate purchase events.
2. Recover the answer turn via **dialogue_pair / next_turn**.
3. Use owner/speaker certificate to reject Speaker A’s preference as the answer.

Preserves who asked, who answered, and who owns the fact.

Why it helps: post-retrieval logic is not a collection of topic-specific answers. It is a small evidence algebra: latest state, add/remove, distinct set, temporal order, date difference, lifecycle filter, and exact-entity mismatch.

HNMW makes the search space small; Query IR makes the evidence complete



LoCoMo evidence fully present	Evidence partially present	No supporting evidence
90.81%	**81.44%**	**55.70%**
answer accuracy	**−9.37 pp**	**−35.11 pp**

These support-stratified results motivate completeness-aware navigation; they are not a standalone Query IR ablation.

The design objective is not “retrieve more.” It is “prove the answer with the smallest source-complete subgraph.”

TAKEAWAY

From retrieval-augmented memory to a queryable memory system

GraphMem turns long history into a persistent world, then compiles each question into a proof obligation.

That combination delivers stronger accuracy while keeping answer-time context bounded.

What is established

- 89.00% on LongMemEval vs Mem0 70.20%
- 86.23% on LoCoMo scored types vs Mem0 78.90%
- 7.03K / 4.41K answer tokens per question
- Source-traceable facts and typed relations

What remains to prove

- Instrument Mem0 build/search/answer token usage
- Run fixed-index module ablations
- Close LoCoMo Temporal and Open-domain gaps
- Reduce first-build cost without losing coverage